a Relational Database Entity-Relationship Architecture (All 12 Tables in 3NF)

TEAMS (48)

PK team_id
fifa_code
confederation
elo_rating

VENUES (16)

PK venue_id
stadium_name
capacity
elevation_m

STAGES (7)

PK stage_id
stage_name
is_knockout

REFEREES (28)

PK referee_id
country
avg_cards_game

MATCHES (104)

PK match_id
FKs: stage, venue, team
home/away_score, xG
referee_id, potm_id

PLAYERS (1,248)

PK player_id
FK team_id
position, height_cm
market_val_eur

EVENTS (601)

PK event_id
FK match, team, player
minute, event_type
(294 Goals, 14 OGs)

LINEUPS (5,408)

PK lineup_id
FK match, player, team
is_starting_xi
minutes_played

TEAM_STATS (208)

PK,FK match_id, team_id
possession_pct
total_shots, saves

PLAYER_STATS (1,248)

PK,FK player_id, team_id
goals (294), assists
clean_sheets, saves

DETAILED (104)

PK match_id
date, home/away_team
score, xG, venue

FEATURES (104)

PK match_id
60 Pre-Match Features
TARGET: match_result
(Zero Data Leakage)

Enforced Engine Integrity: PRAGMA foreign_keys = ON | PRAGMA foreign_key_check = 0 | Compiled Views: vw_match_summaries, vw_team_goal_summary

b Data Processing Pipeline

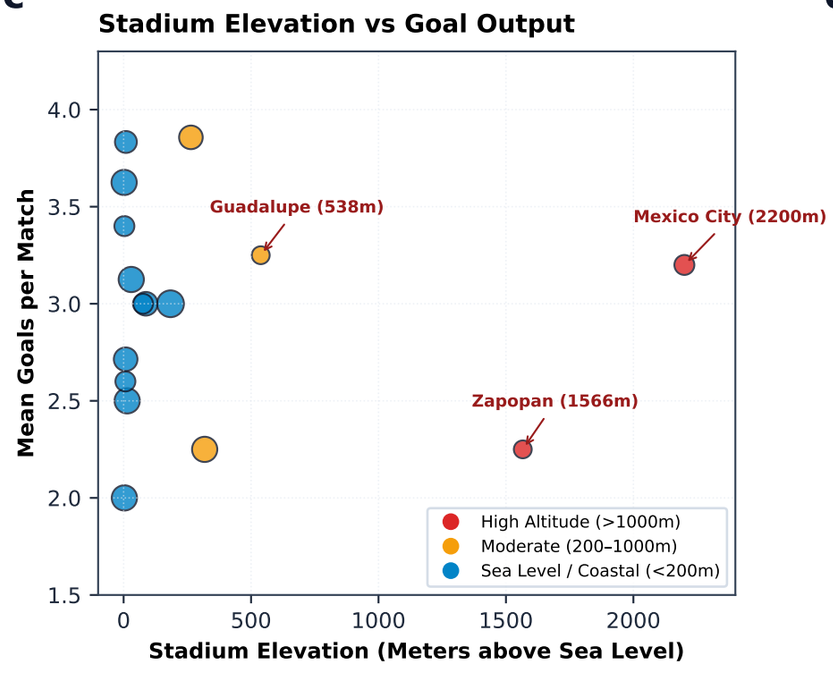
1. Raw Ingestion
4 Public Feeds • (FIFA, Sofascore, • Transfermarkt, GIS)

2. ETL & Normalization
Disambiguation, 3NF • 27 Anomalies Fixed • (1.00% Error Rate)

3. Relational Parity
SQLite Engine, • Snappy Parquet, • RFC 4180 CSVs

4. Predictive Modeling
Zero-Leakage ML • Temporal Train/Test • (M1-72 -> M73-104)

c



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